MILITARY SPECIFICATION

PROPELLANT M26 (T28) FOR USE IN INCREMENT XM90E1 FOR 81MM MORTAR

1. SCOPE

1.1 This specification covers the M26 (T28) Propellant for two types of increments designated as increment XM90E1, A and B.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-G-155 — Graphite

MIL-P-156 — Potassium Nitrate

MIL-B-162 — Barium Nitrate, Technical

MIL-N-244 — Nitrocellulose (For Use in Explosive)

MIL-N-246 — Nitroglycerin

MIL-E-255 — Ethyl Centralite (Carbamite)

MIL-A-2550 — Ammunition and Special Weapons, General Specification for

MIL-I-45208 — Inspection Requirements, General Specification for

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-109 — Inspection Terms and Definitions

MIL-STD-286 — Propellant, Solid; Sampling, Examination and Testing

MIL-STD-1235 — Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ORDNANCE CORPS

76-4-46 — Box, Packing with Metal Line, M24 for Smokeless Powder, Assembly and Details

76-4-56 — Box, Packing Metal-wood, M-17 for Smokeless Powder Assembly

FSC 1375

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8858848 — Marking Diagram and Sealing of Metal Line Wooden Packing Boxes for Shipment of Propellants

CXP-132551 — Charge A, Propellant Increment for XM-90E1 Propellant Charge for 81MM Mortar Loading Assembly

CXP-132562 — Charge A, Bag

CXP-133495 — Charge B, Propellant Increment for XM-90E1 Propellant Charge for 81MM Mortar Loading Assembly

CXP-133496 — Charge B, Bag

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-90—Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations (1949 Edition and revisions) available from the Superintendent of Documents, Government Printing Office, Washington 25, D.C. Orders for the above publications should cite "49 CFR 71-90 (latest revisions)".)

3. REQUIREMENTS

3.1 **Material.** The constituent materials shall comply with the requirements of the applicable specification as follows:

Constituent material	*Conforming to specification*	
Nitrocellulose	MIL-N-244	Grade C Composition I
Potassium Nitrate	MIL-P-156	Class C
Ethyl Centralite	MIL-E-255	Class 2
Nitroglycerin	MIL-N-246	Type I
Graphite	MIL-G-155	Grade II or IV
Barium Nitrate	MIL-B-162	Class 3

3.2 **Composition.** The composition exclusive of total volatile, shall be as specified in table I when determined as specified in 4.3.1.

TABLE I

Ingredient	*Percent*
Nitrocellulose	67.25 plus or minus 1.80
Nitroglycerin	25.00 plus or minus 0.75
Barium Nitrate	0.75 plus or minus 0.20
Potassium Nitrate	0.70 plus or minus 0.25
Ethyl Centralite	6.00 plus or minus 0.50
Graphite	0.30 plus or minus 0.10

3.3 **Total volatiles.** The total volatiles shall not exceed 0.70 percent when tested as specified in 4.3.2.

3.4 **Heat test 120 degrees Centigrade.** The propellant shall not cause complete fading of the methyl violet paper to a salmon pink color in less than 40 minutes, when determined as specified in 4.3.3.

3.5 **Form and dimensions.**

3.5.1 *Form.* The propellant shall be solid disks.

3.5.2 *Dimensions.* The dimensions shall be in accordance with drawing (dwg.) CXP-132551 and CXP-133495.

3.6 **Ballistics.**

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3.6.1 *Velocity.* At 70° F., a weight of propellant assessed as required to impart a service velocity of 269 meters per second (m.p.s.) with a 9 increment charge in M29 Mortar with XM374E1 Cartridge shall not produce a maximum rate pressure of more than 9,200 pounds per square inch (psi) (dynamic pressure).

3.6.2 The lot standard deviation shall not exceed 3.08 m.p.s.

3.6.3 *Individual maximum pressure.* The pressure from any excess charge round or any service condition shall not exceed 115 percent of the dynamic pressure requirement. The excess charge is defined as 150 percent of the weight of 9 increments.

3.7 **Workmanship.** The best commercial practices shall be used in the formulation of propellant furnished under this specification, and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes.

4. QUALITY ASSURANCE PROVISIONS

4.1 **General quality assurance provisions.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL–STD–109 in order to define terms used herein. The provisions of Specification MIL–A–2550 shall apply.

4.1.1 *Contractor quality assurance system.* The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL–I–45208.

4.1.2 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

(b) Number of units of product inspected.

(c) Results obtained for all inspection performed.

(d) Drawing, specification number and date, together with an identification and date of changes.

(e) Certificates of analyses on all material procured directly by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.

(f) Number of items in the lot.

(g) Date submitted

(h) Eight copies of Propellant description sheet, 00 Form 1204, containing the applicable data.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 *Government verification.* Using the

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contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL–I–45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 **Inspection provisions.**

4.2.1 *Lot formation.* A lot shall consist of one or more batches of (T28) propellant produced by one manufacturer, in accordance with the same specification revision, and drawing revisions, under one continuous set of operating conditions. Each batch shall consist of that quantity of (T28) propellant which has been subjected to the same unit chemical or physical mixing process intended to make the final product homogenous.

4.2.2 *Examination.* Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL–STD–105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL–STD–1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect.

4.2.2.1 *Drum sealed.*

Categories	*Defects*	*Method of inspection*	*Code No. (see 6.3)*
Critical: None defined.			
Major: None defined.			
Minor:	AQL 1.00 percent		
201.	Drum improperly sealed	Visual	01001
202.	Marking misleading or Unidentifiable	Visual	01002

4.2.3 *Testing.*

4.2.3.1 Chemical composition, volatiles and stability (see 3.2, 3.3 and 3.4). Major defect. Six pounds of propellant shall be selected at random from the lot. One pound shall be used for chemical, volatiles and stability test. If the sample fails to comply with the requirements specified the lot shall be rejected.

(a)	Chemical Composition	Code No.
	1. Nitrocellulose	02001
	2. Nitroglycerin	02002
	3. Potassium Nitrate	02003
	4. Barium Nitrate	02004
	5. Ethyl Centralite	02005
	6. Graphite	02006
(b)	Total volatiles	03001
(c)	Heat Test	04001

The remaining five pounds shall be forwarded to the Commanding Officer, Picatinny Arsenal, Dover, New Jersey, for 65.5 degree Centigrade surveillance tests. All samples shall be packed in airtight containers and shall be marked to show the propellant designation, lot number, manufacturer, date of sampling, contract number and number of pounds in the lot (see 6.4).

4.2.3.2 Form and dimension (see 3.5) major defect. Thirty disk shall be selected

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from each lot for this examination. If the sample fails to meet the requirements specified the lot shall be rejected.

(a) Form	05001
(b) Diameter	05002
(c) Length	05003

4.2.3.3 Ballistics (see 3.6).

4.2.3.3.1 *Proving ground, major defect.* A ballistic test quantity of 10 pounds for assembling of 10 charges at 9 increments shall be selected from the lot for charge establishment and uniformity at ambient temperature. Any further testing desired as requested by the appropriate government agency, such as at temperature extremes shall be separately provided for by assembly of five charges at each zone of test for each extreme temperature level.

(a) Velocity	06001
(b) Pressure	07001

4.2.3.3.2 *Uniformity of velocity, major defect.* The lot of propellant shall be rejected if the sample Standard Deviation (see 6.2) when multiplied by the factor 0.73 exceeds the applicable requirement. Code No. 08001.

4.2.3.3.3 *Individual maximum pressure, critical defect.* The lot of propellant shall be rejected if any individual round fails to comply with the applicable requirement. Code No. 09001.

4.2.3.4 *Retest.* If for any reason the proof director considers that the test conditions have detrimentally affected the test results, he shall declare the results invalid and perform a new test on additional charges as required.

4.3 Test methods and procedures.

4.3.1 *Composition.* The composition shall be determined on a graphite (glazed) and total volatiles-free basis.

4.3.1.1 *Nitroglycerin.* The percentage of nitroglycerin shall be determined in accordance with Standard MIL-STD-286, method 208.1 using methylene chloride as the solvent for extraction

4.3.1.2 *Ethyl centralite.* The percentage of ethyl centralite shall be determined as specified in Standard MIL-STD-286, method 202.2.

4.3.1.3 *Potassium nitrate.* The percentage of potassium nitrate shall be determined in accordance with Standard MIL-STD-286, method 310.4.

4.3.1.4 *Graphite.* The percentage of graphite shall be determined in accordance with Standard MIL-STD-286, method 308.1.

4.3.1.5 *Nitrocellulose.* The percentage of nitrocellulose shall be determined in accordance with Standard MIL-STD-286, method 209.2.

4.3.1.6 *Barium Nitrate.* The percentage of barium nitrate shall be determined in accordance with Standard MIL-STD-286, method 304.1.

4.3.2 *Total volatiles.* The percentage of total volatiles shall be determined in accordance with Standard MIL-STD-286, method 103.3.

4.3.3 *Heat test, 120 degrees Centigrade.* The 120 degrees Centigrade heat test shall be conducted in accordance with Standard MIL-STD-286, method 404.1.

4.3.4 *Form and dimensions.*

4.3.4.1 *Form.* The propellant shall be examined visually for form.

4.3.4.2 *Dimensions.* The dimensions shall be measured using a magnifying instrument, such as a microscope or optical comparator, giving suitable magnification (20-50 X is satisfactory) and equipped with cross hairs, horizontal scale, and vernier screw scale capable of measurement to at least 0.001 inch.

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4.3.5 *Ballistics.*

4.3.5.1 *Charge establishment.* Increment charge weights shall be determined by firing propelling charge assembled in accordance with drawing 8881021 and 8881023 (or closed bomb firing when requested) to meet the requirements specified in 3.6.

5. PREPARATION FOR DELIVERY

5.1 Packing. The powder shall be packed in metal drums. The type shall be in accordance with dwg. 76–4–46 and 76–4–56 or equivalent drum of equal quality so constructed to meet ICC regulation and to insure acceptance by common or other carrier for safe transportation at the lowest rate to the point of delivery. Immediately prior to packing all containers shall be subjected to an internal pressure of ½ to 1 pound per square inch by a method satisfactory to the contracting office's representative. A water manometer shall be assembled in the system. A drop of 0.7 inches or more on the manometer in 15 seconds shall be cause for rejection and the container shall be removed from the lot.

5.2 Marking. In addition to any marking required by the contract or order marking shall be in accordance with dwg. 8858848 and Code of Federal Regulations "49 CFR 71–90."

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Process. Details of the manufacturing process and the equipment used by the contractor will be submitted to the procuring activity in writing prior to commencement of manufacture. Any deviation from this manufacturing process must be submitted in writing to the procuring activity prior to being put into effect.

(c) Form of propellant required (see 3.6).

6.2 Standard deviation. The Standard deviation will be calculated with (N–1) as the divisor in a standard statistical technique equivalent of that shown in Standard MIL–STD–414, Section B, Example B–1.

6.3 Inspection code numbers. The five-digit code numbers assigned to the inspection herein are to facilitate future data collection and analysis by the Government.

6.4 Surveillance sample. The expense of surveillance testing shall be borne by the Government. All samples will be packed in air-tight containers and will be marked to show the propellant designation the projectile for which manufactured, lot number, manufacturer, date of sampling, contract number and number of pounds in the lot.

6.5 Submission of test data. In addition to the normal distribution of records, when the propellant is procured by the Department of the Army, one copy of ballistic data will be forwarded to the following:

Commanding Officer
Picatinny Arsenal
ATTN: SMUPA–ND2
Dover, N. J. 07801

Custodian:
Army—MU

Preparing activity:
Army—MU

Project No. 1375–A649

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